

**PILOT STUDY**  
**Antidepressant effects on natural products: systematic review**

1. **Title:** Antidepressant-like effects and molecular targets of plant-derived compounds in rodent models: a systematic review and meta-analysis.
2. **Authors:** Fatima C Hamze, Juliana A Bolzan, Maria C S Bittencourt, Cilene L de Oliveira.
3. **Citation:** Hamze, F.C.; Bolzan, J.A.; Bittencourt, M.C.S.; Lino de Oliveira, C. Antidepressant effects on natural products: systematic review. September, 2026. Pilot study.
4. **Databases:** Medline (via PubMed), Scopus, Embase, and Web of Science (WOS).
5. **Date of Search:** 09/21/2026.
6. **Search strategy:**

| PubMed                                           |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Results   |
|--------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| #1                                               | Population<br>MeSH terms,<br>title/abstract<br>filter   | "Rats"[Mesh] OR "Mice"[Mesh] OR rats[tiab] OR rat[tiab] OR mouse[tiab] OR mice[tiab]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4,041,049 |
| #2                                               | Intervention<br>MeSH terms,<br>title/abstract<br>filter | "Plants, Medicinal"[Mesh] OR "Phytotherapy"[Mesh] OR "Plant Extracts"[Mesh] OR "Plant Preparations"[Mesh] OR phytotherapy[tiab] OR "herbal medicine"[tiab] OR "natural product*" [tiab] OR "medicinal plant*" [tiab] OR "plant extract*" [tiab] OR "herbal drug*" [tiab] OR phytochemical* [tiab] OR phytomedicine[tiab] OR "crude extract*" [tiab] OR phytocomplex* [tiab] OR "essential oil*" [tiab] OR botanical* [tiab]                                                                                                                                                                                                      | 534,934   |
| #3                                               | Outcome<br>MeSH terms,<br>title/abstract<br>filter      | "Depressive Disorder"[Mesh] OR antidepress* [tiab] OR "depressive-like behavior" [tiab] OR "depressive-like behaviour" [tiab] OR "chronic mild stress" [tiab] OR "chronic unpredictable stress" [tiab] OR "forced swim test" [tiab] OR "forced swimming test" [tiab] OR "forced swimming tests" [tiab] OR "Porsolt test" [tiab] OR "Porsolt tests" [tiab] OR "tail suspension test" [tiab] OR "learned helplessness" [tiab] OR "sucrose preference test" [tiab] OR "splash test" [tiab] OR "novelty suppressed feeding" [tiab] OR "novelty-induced hypophagia" [tiab] OR "behavioral despair" [tiab] OR "immobility time" [tiab] | 212,814   |
| #1 AND #2 AND #3 (filter applied: other animals) |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2,173     |

| Web of Science |                               |                                                                                                                                                                                                                                                               | Results   |
|----------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| #1             | Population<br>Topics filter   | TS=(rat OR rats OR mouse OR mice)                                                                                                                                                                                                                             | 4,223,735 |
| #2             | Intervention<br>Topics filter | TS=("medicinal plant*" OR phytotherapy* OR "plant extract*" OR "plant preparation*" OR "herbal medicine*" OR "natural product*" OR "herbal drug*" OR phytochemical* OR phytomedicine* OR "crude extract*" OR phytocomplex* OR "essential oil*" OR botanical*) | 484,832   |

|                                                   |                                 |                                                                                                                                                                                                                                                                                                                                                                                                 |         |
|---------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| #2                                                | <b>Outcome</b><br>Topics filter | TS=("depressive disorder*" OR antidepress* OR "depressive-like behav*" OR "chronic mild stress" OR "chronic unpredictable stress" OR "forced swim* test*" OR "Porsolt test*" OR "tail suspension test*" OR "learned helplessness" OR "sucrose preference test*" OR "splash test*" OR "novelty suppressed feeding" OR "novelty-induced hypophagia" OR "behavioral despair" OR "immobility time") | 190,586 |
| <b>#1 AND #2 AND #3</b> (filter applied: article) |                                 |                                                                                                                                                                                                                                                                                                                                                                                                 | 1,141   |

| Scopus                                            |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                            |           |
|---------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                                                   |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                            | Results   |
| #1                                                | <b>Population</b><br>Title/abstract/keywords filter   | TITLE-ABS-KEY (rat OR rats OR mouse OR mice)                                                                                                                                                                                                                                                                                                                                                               | 4,748,417 |
| #2                                                | <b>Intervention</b><br>Title/abstract/keywords filter | TITLE-ABS-KEY ("medicinal plant*" OR phytotherapy* OR "plant extract*" OR "plant preparation*" OR "herbal medicine*" OR "natural product*" OR "herbal drug*" OR phytochemical* OR phytomedicine* OR "crude extract*" OR phytocomplex* OR "essential oil*" OR botanical*)                                                                                                                                   | 818,216   |
| #2                                                | <b>Outcome</b><br>Title/abstract/keywords filter      | TITLE-ABS-KEY ("depressive disorder*" OR antidepress* OR "depressive-like behav*" OR "chronic mild stress" OR "chronic unpredictable stress" OR "forced swim* test*" OR "Porsolt test*" OR "tail suspension test*" OR "learned helplessness" OR "sucrose preference test*" OR "splash test*" OR "novelty suppressed feeding" OR "novelty-induced hypophagia" OR "behavioral despair" OR "immobility time") | 330,814   |
| <b>#1 AND #2 AND #3</b> (filter applied: article) |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                            | 3,123     |

| Embase                                            |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |
|---------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                                                   |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                             | Results   |
| #1                                                | <b>Population</b><br>Emtree terms, title/abstract/keywords filter   | ('rat'/exp OR 'mouse'/exp OR (rat OR rats OR mouse OR mice):ti,ab,kw)                                                                                                                                                                                                                                                                                                                                                       | 5,001,119 |
| #2                                                | <b>Intervention</b><br>Emtree terms, title/abstract/keywords filter | ('medicinal plant'/exp OR 'phytotherapy'/exp OR 'plant extract'/exp OR ("medicinal plant*" OR phytotherapy* OR "plant extract*" OR "biological product*" OR "plant preparation*" OR "herbal medicine*" OR "natural product*" OR "herbal drug*" OR phytochemical* OR phytomedicine* OR "crude extract*" OR phytocomplex* OR "essential oil*" OR botanical*):ti,ab,kw)                                                        | 775,155   |
| #2                                                | <b>Outcome</b><br>Emtree terms, title/abstract/keywords filter      | ('depression'/exp OR ("depressive disorder*" OR antidepress* OR "depressive-like behav*" OR "chronic mild stress" OR "chronic unpredictable stress" OR "forced swim* test*" OR "Porsolt test*" OR "tail suspension test*" OR "learned helplessness" OR "sucrose preference test*" OR "splash test*" OR "novelty suppressed feeding" OR "novelty-induced hypophagia" OR "behavioral despair" OR "immobility time"):ti,ab,kw) | 931,021   |
| <b>#1 AND #2 AND #3</b> (filter applied: article) |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                             | 3,602     |

7. **Terms for search in databases:** the search for #1 (terms for population) AND #2 (terms for intervention) AND #3 (terms for outcome) were selected for this pilot study using “title or abstract” or “title, abstract or keywords” and/or MeSH/Emtree terms as filters in the databases.
8. **Results search:** 10,039 publications on September 21, 2026.
9. **Reference manager:** EndNote will be used.
10. **References:**
  - a. Bolzan JA, Lino de Oliveira C. Protocol for systematic review and meta-analysis of the evidence linking hippocampal neurogenesis to the effects of antidepressants on mood and behavior. *BMJ Open Sci.* 2021 Mar. 4;5(1):e100077.
  - b. Ramos-Hryb AB, Bahor Z, McCann S, et al. Protocol for a systematic review and meta-analysis of data from preclinical studies employing the forced swimming test: an update. *BMJ Open Science* 2019;3:e000043.
  - c. Bolzan, J.A., Lino de Oliveira, C. (2020, January 31). Pilot studies.